MATH 111 Calculus 1

The derivative

Tangents, velocity, and rates of change

Week 3 · Lecture 7 · Stewart §2.1

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. In the meantime: a stone is dropped from the roof of a 45-metre building, and after t seconds it is $45 - 5t^2$ metres above the ground. How fast is it going *at the instant it hits the ground*? Not on average over the fall — at that instant.

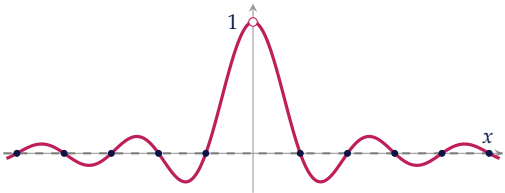
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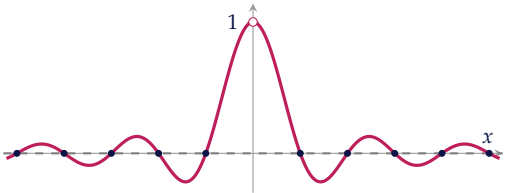


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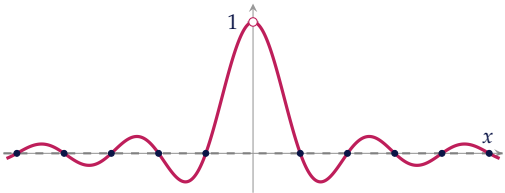
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"Two weeks on limits. When do we use them for something?"

Today. Every question in this lecture — how steep is this curve, how fast is this stone — turns out to be *one limit*, and it is the same limit every time.

STAND UP

"A tangent line touches the curve at exactly one point."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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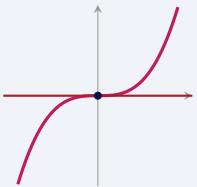
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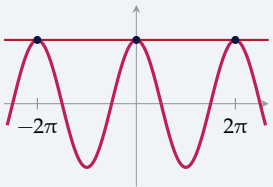
Verdict

FALSE. That is the *circle*'s tangent, learned at school. The tangent to $y = x^3$ at the origin is $y = 0$, and it crosses the curve there; the tangent to $y = \cos x$ at $x = 0$ is $y = 1$, and it touches the curve again every 2π. Touching once is not the point; having the same direction is. Today we say that with a limit.

$y = x^3$: it crosses

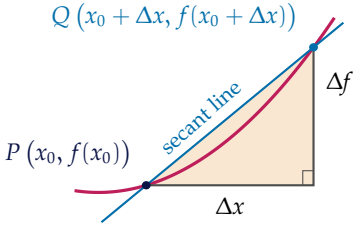


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Geometric definition of the derivative

A tangent line is *not* simply a line that meets the graph once. It is the limit of the secant lines through P and a second point Q of the graph, as the distance between P and Q goes to zero.

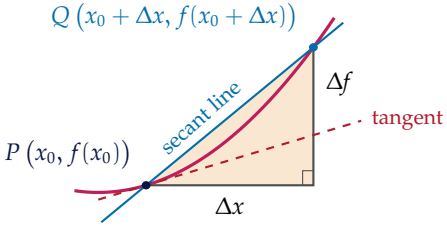


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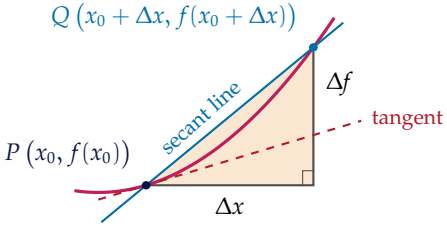


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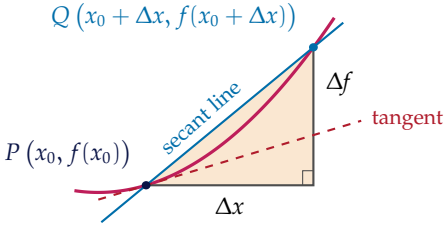
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$$\lim_{\Delta x \to 0} \underbrace{\frac{\Delta f}{\Delta x}}_{\text{difference quotient}} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \underbrace{f'(x_0)}_{\text{derivative of } f \text{ at } x_0}$$

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With $x = x_0 + \Delta x$ the same limit is $\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$. When the limit exists, the tangent line at P is the line through P with slope $f'(x_0)$.

Notation — several names for one number

Calculus was built by several people, and — like a language — it kept several ways of saying the same thing. With $y = f(x)$, the change in y ("delta y", "delta f") is

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Say where — and one more name

Leibniz's notation hides the point: write $\left.\frac{dy}{dx}\right|_{x=x_0} = f'(x_0)$. Many books write h for Δx: $f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$. Also: $\frac{df}{dx}$, f', Df.

A tangent, computed

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Recall: line through (x_0, y_0), slope m: $y - y_0 = m(x - x_0)$

With $x_0 = 2$, $y_0 = 5$, $m = 4$: the tangent is $y - 5 = 4(x - 2)$, that is $y = 4x - 3$.

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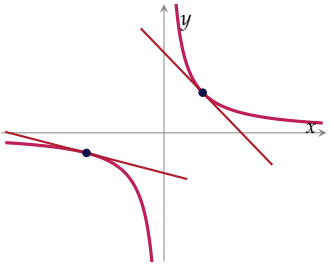
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The shape of every such computation

Write the difference quotient; substitution gives $0/0$; rewrite until the offending factor cancels; *then* take the limit. Week 1's algebra, week 3's meaning.

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It is tempting to put $\Delta x = 0$ right away — but then you always get $0/0$. You always have to cancel first.

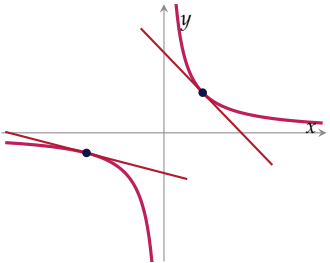


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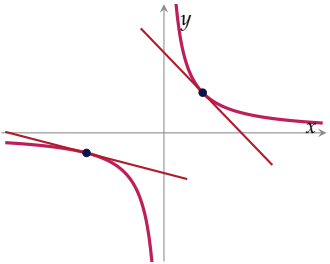
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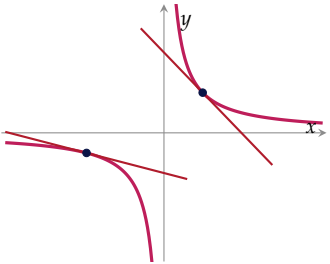
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Taking the limit as $\Delta x \to 0$: $\displaystyle\lim_{\Delta x \to 0} \frac{-1}{(x_0 + \Delta x)\, x_0} = \frac{-1}{x_0^2}$.

Hence $f'(x_0) = -\frac{1}{x_0^2}$.



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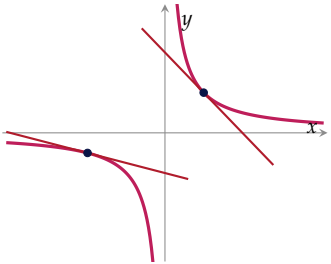
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Notice that $f'(x_0)$ is negative — as is the slope of the tangent line on the graph.

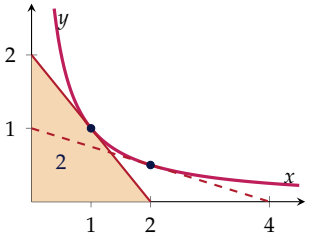


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What the tangent cuts off — a gift from the hyperbola

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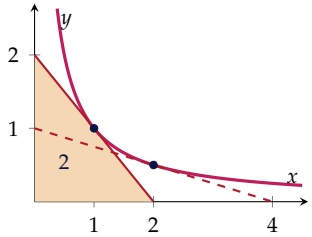


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Where it meets the axes. x-intercept: put $y = 0$; then $-\frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0)$, so $x - x_0 = x_0$ and $x = 2x_0$.
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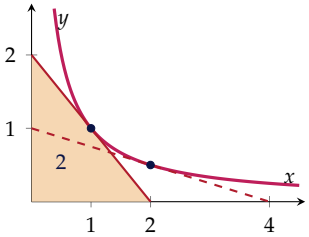
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The area of the triangle

Area $= \frac{1}{2}(2y_0)(2x_0) = 2x_0y_0 = 2x_0\left(\frac{1}{x_0}\right) = 2$ — always 2, no matter where on the graph the tangent line is drawn.



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Question 14

Which of these gives the slope of the tangent to $y = f(x)$ at $x = 4$?

A. $\lim_{x \to 4} \dfrac{f(x) - f(4)}{x - 4}$

B. $\dfrac{f(x) - f(4)}{x - 4}$

C. $\lim_{h \to 0} \dfrac{f(4 + h) - f(4)}{h}$

D. Both A and C.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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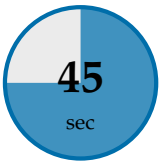
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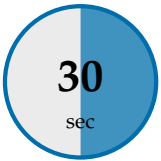
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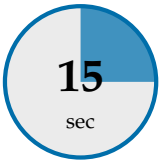
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A B C D

Question 14

EXPLAIN

Which of these gives the slope of the tangent to $y = f(x)$ at $x = 4$?

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

D. A and C are the same number written with two different letters: put $h = x - 4$ and one becomes the other. B is the trap — it is the slope of a *secant*, a whole family of numbers depending on x, and no limit has been taken. The word "tangent" is exactly the word "lim".

STAND UP

"Velocity is distance divided by time."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

FALSE as stated. Distance over time is the average velocity over an interval. A speed camera does not average your trip; it reports how fast you were going *at one instant* — and that number is again a limit, of average velocities over shorter and shorter intervals. Same limit as the tangent slope, wearing different clothes.

Question 15

Back to the title slide. A stone is dropped from the roof of a 45-metre building; after t seconds it is $y(t) = 45 - 5t^2$ metres above the ground. It lands at $t = 3$, so its *average* speed over the fall is $45/3 = 15$ m/s. How fast is it going at the instant it hits the ground?

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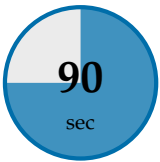
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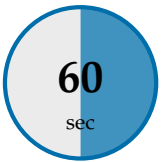
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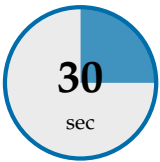
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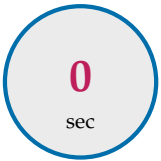
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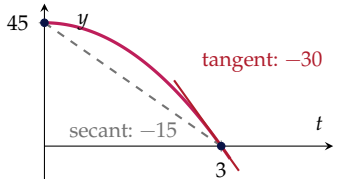
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Answer

B. The speed at the instant of impact is the limit of average speeds over shorter and shorter intervals ending at $t = 3$: $\frac{y(3+h)-y(3)}{h} = -30 - 5h \to -30$. So 30 m/s downwards — *twice* the average. A is the average itself; C is the height of the roof, not a speed; D is exactly what the limit is for. The next slide draws it.

Back to the title slide: from average to instantaneous

Over $[t_0, t_0 + h]$, $h = \Delta t$, the average velocity $\dfrac{\Delta y}{\Delta t} = \dfrac{y(t_0 + h) - y(t_0)}{h}$ is a *secant* slope; its limit as $h \to 0$ is the instantaneous velocity $v(t_0)$, a *tangent* slope.

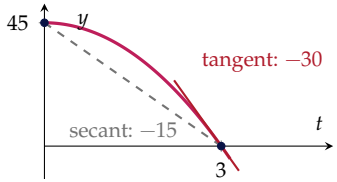


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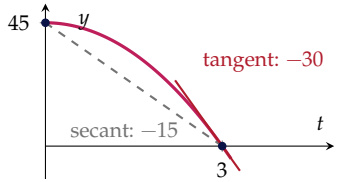
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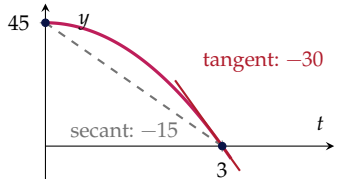
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$v(3) = \lim_{h \to 0}(-30 - 5h) = -30$ m/s: twice the average, and negative — downwards. The title slide's answer: 30 m/s at impact.

One definition for both — the derivative

Definition

The derivative of f at x_0 is

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

if this limit exists. Equivalently $f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$.

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If the limit does not exist, f has no derivative at x_0 — Lecture 8 is about how that happens. In Leibniz's notation (slide 5) the same number is dy/dx.

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Step 4. Let $h \to 0$: $f'(2) = \lim_{h \to 0}(h - 6) = -6$.

The tangent at $(2, -9)$ is $y + 9 = -6(x - 2)$. Negative slope: at $x = 2$ the curve is going *down*.

Question 16

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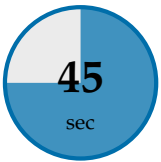
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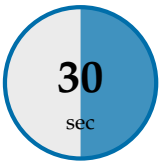
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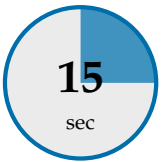
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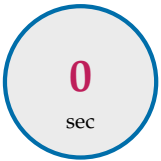
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Answer

C. A derivative is a rate, so it carries units: dollars *per metre*. It compares a small change in the output with the small change in the input that caused it — which is what "one more metre costs about 14" says. A confuses the rate with the value $C(60)$; B is the average $C(60)/60$, a different number; D invents a variable (t) that is not in the problem.

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Reading a derivative aloud

$f'(x_0) = k$ says: near x_0, one more unit of x changes y by about k units — the sign gives the direction, the size the speed, and the units are "units of y per unit of x".

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Notice what happened: the trick that rescued a limit in week 1 is the trick that finds a tangent in week 3. Nothing new was needed — only a new question.

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What neither had was the limit. That came from **Cauchy** in 1823 and **Weierstrass** in the 1860s — the ε and δ you learned in week 1 are the last brick, laid two hundred years after the house was built.

Today in three lines

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This week:

Homework: Stewart §2.1 — read and work out the examples. Next: the derivative as a function.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"The derivative of f at x_0 is …" (in your own words, and say what it measures).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.